

Machine Learning Final Project

Bonus Track Report

This report is for the final project bonus track of Prof. Hsuan-Tien Lin's Machine Learning course in 2023 Fall. This report delves into the debate among artificial intelligence agents, focusing on the impact of adjusting various parameters and prompts on the debate outcomes. Inspired by SocraSynth, our role as moderators becomes crucial, aiming to guide the conversation and explore improvements by adjusting experimental parameters. Our study involves practical scenarios with 80 debate topics and a CRIT scoring system to enhance agent-generated conclusions and CRIT scores. Our goal is to systematically study how to configure the parameters and prompts to reach the best score. Finally, our conclusion is that the parameters do not significantly affect the debating process. Moreover, requesting the presentation of data sources, and providing agents with historical examples and scores can contribute to achieving the best configuration.

I Introduction

In this final report, we delve into the debate among artificial intelligence agents, focusing on the impact of adjusting various parameters and prompts on the debate outcomes. Our research is inspired by SocraSynth, a platform that integrates the principles of "Socratic Synthesis" and "Socratic Symposium," aiming to foster constructive dialogue and critical thinking through simulated dialogues and interactions.

In terms of structure, this report unfolds systematically. The first chapter serves as an introduction, providing a comprehensive overview. Subsequent chapters include Auto Program, introducing the automated program used in our experiments; Method, explaining the three proposed experiments along with the corresponding rationale and identified areas worthy of experimentation; Experiment Results, presenting our experimental findings; Combine all method in the fifth chapter, where we observe the benefits of various methods and propose combining them for further experimentation; and the final chapter, Conclusion, where we present our ultimate recommended configuration for addressing the research question. Additionally, we provide an appendix containing the complete prompt.

II Auto program

In our project, we have introduced an automated program designed to facilitate the process of thematic debates. The operation of this automation is based on a series of codes utilizing tools such as **Selenium** to simulate browser actions, automate login processes, and submit configurations. Through the utilization of this program, we can facilitate dynamic interactions with two distinct roles, engaging in conversations with agents that represent opposing positions on a given topic. This automated debating process covers various stages of debate, including topic selection, position expression, dialogue exchange, and ultimately, conclusion and summarization.

This functionality allows us to conduct thematic debates more rapidly and objectively while ensuring fairness for each role throughout the debate process. The crucial data generated by this program significantly enhances the efficiency of debates, providing robust support for the overall progress and discussions within the

project.

In our program, we initially retrieve all the topics and then use a loop to allow the program to debate each topic sequentially. This structure ensures comprehensive debate coverage for all topics, simultaneously enhancing the program's efficiency.

Now, we present a glimpse into our code:

```
# submit config
textarea = driver.find_element_by_id('llm_config')
textarea.send_keys(config_data)
form = driver.find_element_by_tag_name('form')
form.submit()
```

This section involves submitting the model configuration, and other operations, such as engaging in conversations with agents, follow a similar approach. After these process, we will enter the chatUI playground where we can talk with two agents.

To talk with the two agents, we prepared the prompts for each stage of the debate. And using the below example code to chat with agent-A and agent-B:

```
select.select_by_visible_text("Agent-A")
talk(to_a, submit_button, user_input_element)
#to_a : a prompt to agent-A
robot_response_A = store_talk_response('agentAmessage')
#robot_response_A : store what A said
```

Now, we can chat with two agents automatically. However, we found that the agents' speaking speed is too slow, and it will take us approximate 1-2 hours two complete one debate. We found the reason for that is in the source code of the website:

```
336 iv>
337
338 temp = document.getElementById('temp')
339 index = 0
340 ction displayNextLetter() {
341   scrollToBottom()
342   if (index < message.length) {
343     temp.innerHTML = temp.innerHTML + message[index++];
344     setTimeout(displayNextLetter, 30);
345   } else {
346     temp.removeAttribute('id')
347     sendbtn.disabled = false
348     userInputarea.disabled = false
349   }
350 }
351
352 playNextLetter()
353 ollToBottom()
354
```

Figure 1: source code of website

The setTimeout function determine the speed of the agents' talking speed, our target is to change the param-

eter '30' to a lower number. Therefore, there is a distinct process in our code right after we enter the playground:

```
change_time_out_function = """
var originalSetTimeout = window.setTimeout;
window.setTimeout = function(callback, time) {
  if (
    callback.toString().indexOf('displayNextLetter') == 1
    || callback.toString().includes('temp.innerHTML')) {
    time = 1; }
  return originalSetTimeout(callback, time);
};
"""
driver.execute_script(change_time_out_function)
```

This adjustment reduces the delay in `setTimeout(callback, delay)`, thus expediting the process.

III Baseline

To set a baseline score for the project, we referenced the debate procedure and prompts by the TA [1]. In brief, as moderators, we initially present the subject to the agents, guide the two agents in selecting five topics, facilitate their debate (debate goes back and forth 3 times), and subsequently prompt them for a conclusive summary. After running through all 40 public subjects, the scores are shown in Fig. 2.

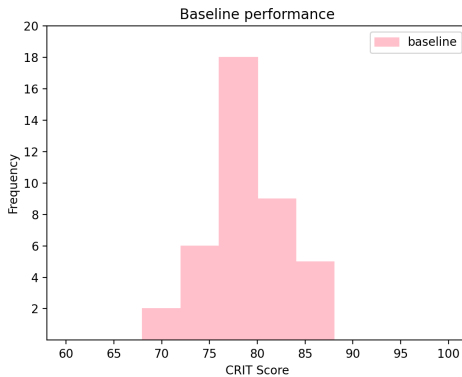


Figure 2: Baseline performance on all 40 subjects.

IV Method

A Parameter

We referred to the examples provided by the assistant[1], as well as the API reference provided by OpenAI[2], and selected three parameters that should have the most significant impact on the results, namely `"presence_penalty"` and `"top_p"` in the request body, and the `argument_strength` in the prompt. We initially set `"presence_penalty"` to 0.3, `"top_p"` to 0.1, and `argument_strength` to 0.9 as the control group. Subsequently, we adjusted the values of `"presence_penalty"`, `"top_p"`, `argument_strength` separately for experimentation.

B Evidence reference enhancement

Since the main goal of this project is to obtain the highest CRIT score, it is the best to understand the CRIT system by conducting a research on CRIT.

1 Background on CRIT scores

According to Chang’s work [3], the evaluations take place in an empty context. CRIT runs on multiple foundational models, including GPT-4, GPT-3.5, and text-davinci 003. This approach forms a panel of judges with subtly different knowledge base. Then CRIT is asked: I request that you, [GPT4, GPT3.5, daVinci-003], analyze a document using the following five steps:

1. Identify the document’s main claim or conclusion.
2. Locate the arguments that support this main claim.
3. Score the validity of the reasoning or inference for each argument on a scale of 0 to 10 (strong).
4. For each argument, identify counterarguments and score the reasoning validity on the same scale.
5. Determine the winning side, be it Agent A or B, and provide justifications for this decision.

To understand the factors contributing to a favorable CRIT score, we replicated the CRIT simulation using GPT-3.5, employing an identical methodology. We began with asking it how to improve the validity score of an argument in a debate. These are its suggestions:

1. Support with Objective Evidence: Incorporate data, statistics, or research findings to back up your claims. Use reputable sources, such as academic journals, rankings, or official reports, to add credibility to your argument.
2. Diversify Your Evidence: Include a variety of supporting points that cover different aspects of the topic. Consider different perspectives, angles, or dimensions of the issue to provide a more comprehensive view.
3. Expert Opinions: Cite experts in the field who have authority and expertise on the subject. Referencing well-known scholars or professionals can enhance the credibility of your argument.
4. Logical Reasoning: Ensure that your argument follows a logical structure with clear connections between premises and conclusions.

2 CRIT simulation

We first gave the following prompt to our CRIT simulator:

*Score the validity of the reasoning or inference for the argument below on a scale of 0 to 10 (strong):
National Taiwan University is good at physics because I am studying there.*

This argument was given a **Validity Score of 3** out of 10 by the CRIT simulator with the response saying, "This score reflects that personal experience is a limited form of evidence. To improve the validity, you could supplement your argument with additional objective criteria such as academic rankings, faculty expertise, research output, or other external measures that provide a more thorough evaluation of the physics department at National Taiwan University." Next we changed the argument to:

National Taiwan University is good at physics because experts say so.

This input gave us a **Validity Score of 7** out of 10, and the output was: "This score reflects the acknowledgment that expert opinions lend weight to the argument, but it does not reach the highest score because the identity and qualifications of these experts are not specified." As a result, we updated the prompt:

National Taiwan University is good at physics because experts from US governments say so.

This ended up as a **Validity Score of 8** out of 10. Finally, we added one more evidence to support our argument:

National Taiwan University is good at physics according to the QS ranking and because experts from US governments say so.

The score is improved to a **Validity Score of 9** out of 10. Hence, we have shown that a high CRIT score is strongly related to the evidence of your argument. Additionally, having comprehensive information about the source of the evidence enhances the overall quality of the evaluation.

C Giving history example

We thought that ChatGPT currently struggles to generate satisfactory conclusions, and one contributing factor is the lack of reference examples. Therefore, we have decided to provide an additional "history example" to both agents. This involves showing them higher-scoring conclusions from previous evaluations, allowing them to gain insight into how to generate better conclusions. So we provide them the below prompts:

"Agent-A, as the proponent of the subject, you advocate the debate topics, so please provide the conclusions of your argument on the five debate topics. Here are some conclusions from previous debates on different topics. Their debate subject is "... and they received a score of ... for their conclusions. You can refer to them when summarizing conclusions for the your current debate topics:

The example conclusion of history debate will put here

After you provided your conclusion, some AI will score you. The goal is let your conclusion get higher score from the AI."

V Experiment results

A Parameter

We created three figures based on three variables: p, t, and s. These figures illustrate the average scores of different topics under varying values.

In this context, p represents "presence_penalty", t represents "top_p", and s represents "argument_strength".

Each figure corresponds to one of these variables (p, s, or t) and displays a line graph representing the average scores of different topics across different values of that variable.

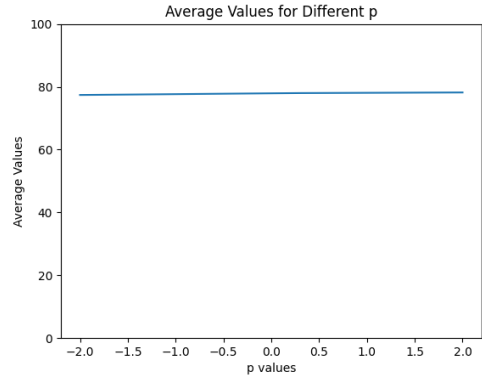


Figure 3: Average Values for Different p

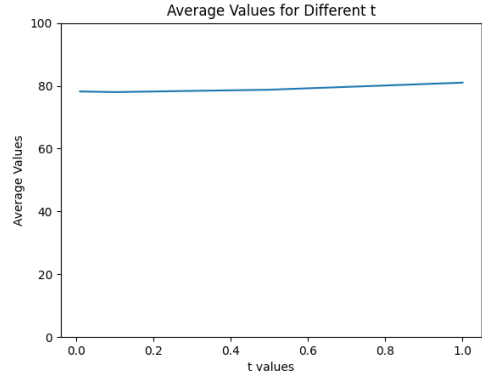


Figure 4: Average Values for Different t

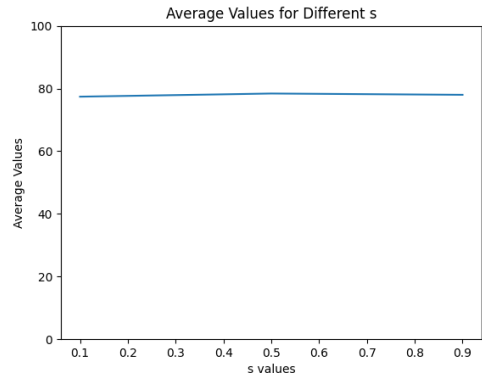


Figure 5: Average Values for Different s

From the results shown in Figures 3, 4, and 5, it can be observed that varying parameter values do not have a significant impact on the outcomes.

To further substantiate our result, we conducted the Kruskal-Wallis test to assess whether different parameter values lead to statistically significant differences in the

Table I: Comparison of p-values under different parameters

Parameter	p	t	s
p-value	0.00674	0.00674	0.00017

Table II: Values of parameters

Parameter name	p	t	s
Parameter value	1	1	0.5

outcomes. The test statistic is given by

$$H = (N - 1) \frac{\sum_{i=1}^g n_i (\bar{r}_i - \bar{r})^2}{\sum_{i=1}^g \sum_{j=1}^{n_i} (r_{ij} - \bar{r})^2}, \quad (1)$$

where N is the total number of observations across all groups, g is the number of groups, n_i is the number of observations in group i , r_{ij} is the rank (among all observations) of observation j from group i , $\bar{r}_i = \frac{\sum_{j=1}^{n_i} r_{ij}}{n_i}$ and $\bar{r} = \frac{1}{2}(N + 1)$.

The obtained p-values are presented in Table IV. The null hypothesis we used is that at least one population is located different from the others in the set. All these p-values are smaller than the chosen significance level of 0.05, indicating that we have sufficient evidence to reject the null hypothesis. In other words, we can infer that different parameter values have no impact on the final performance.

Thus, combining visual observations with the results of statistical tests, we conclude that variations in these parameter values do not appear to have a statistically significant impact on the ultimate performance.

B Implementing evidence reference enhancement

1 Method I: reference enhance

To emphasize the importance of agents providing evidence and citing references to support their arguments, we simplified the topic deciding process, and we updated the following line in the input prompts:

Please prepare evidence to strengthen your argument. Please remember to cite your reference for every evidence.

The parameter used here is shown in Table II.

We have randomly selected five subjects to test our upgraded prompt. As shown in the figure 7, the performance has improved in the selected subjects, and the average of delta score is in Table III.

Table III: Results after implementing evidence reference enhancement

Subject	baseline	new score	delta score
1. cryptocurrency	76	83	+7
2. space exploration	69	81	+12
3. virtual games	81	81	+0
4. chicken eggs	76	85	+9
5. lost penny	76	77	+1

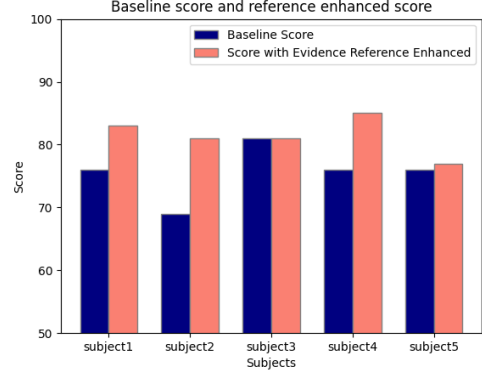


Figure 6: Scores before and after the Evidence Reference Enhancement

2 Method II: include statistical data

To remind the debate agents to include statistical results to support their arguments, we updated the prompt with:

Please provide statistical numbers from previous experiment results in research to support your argument.

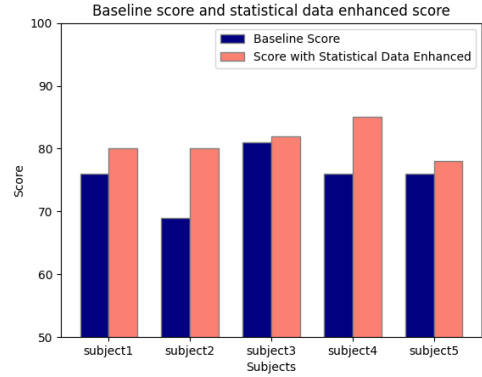


Figure 7: Scores before and after the Statistical Data Enhancement

3 Mann-Whitney U test

Additionally, in this experiment, we employed the Mann-Whitney U test to compare two sets of data which is defined as below:

Definition V.1. Let $X_1, \dots, X_{n_1}$ be an i.i.d. sample from X , and $Y_1, \dots, Y_{n_2}$ an i.i.d. sample from Y , and both samples independent of each other. The corresponding Mann-Whitney "U" statistic is defined as the smaller of:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1, \quad (2)$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 \quad (3)$$

with R_1, R_2 being the sum of the ranks in groups 1 and 2, respectively.

Table IV: Comparison of p-values under different evidence reference enhancement.

Parameter	Method I	Method II
p-value	0.0322	0.0316

The null hypothesis posited that the two populations are equal. Following the test, we obtained a p-value of 0.0322. Since this p-value is less than our chosen significance level, 0.1, we have sufficient evidence to reject the null hypothesis. Therefore, we get the conclusion that there is a significant difference between the two populations, which means the modify exactly improves the final performance.

C Giving history example

The experiment results of given the history example to the two agent is as table VI.

Table V: Result of debate for provide history example

Subject	baseline	new score	delta score
1. cryptocurrency	76	81	+5
2. space exploration	69	77	+8
3. virtual games	81	79	-2
4. chicken eggs	76	83	+7
5. lost penny	76	84	+8

Let's visualize it in a graph for easier observation, the results is in figure8.

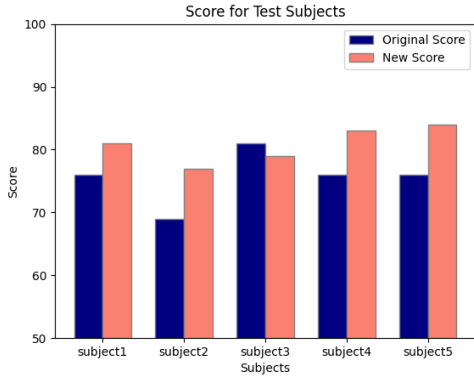


Figure 8: Scores of Test Subjects for providing history example

It is evident from the graph that our approach, which involves providing agents with examples of previous debate results, can effectively improve the quality of debate conclusions.

Additionally, in this experiment, we also employed the Mann-Whitney U test to compare two sets of data. The null hypothesis posited that the two populations are equal. Following the test, we obtained a p-value of

$$p - value = 0.0438. \quad (4)$$

Since this p-value is less than our chosen significance level, we have sufficient evidence to reject the null hypothesis. Therefore, we get the conclusion that there is a

significant difference between the two populations, which means the modify exactly effect the final performance.

VI Combine all method

So far, we have attempted to adjust three aspects of the debate:

- (1) The parameters of ChatGPT (including argument strength and penalty, etc.)
- (2) Requesting the agents to include research data and references.
- (3) Providing agents with historical examples and scores.

We now want to combine all the methods together. First of all, since we found that the parameter do not appear to have a statistically significant impact on the ultimate performance, we choose to use the original one, which is the same when we doing the second and third method as mentioned in Tabel II. In addition, we updated the prompt according to Section V B so that the agents will provide multiple evidence and citations to support their arguments. Finally, as detailed in Section V C, we provide the debating agents with past high-scoring examples as templates, guiding them to formulate responses based on these templates.

And our results, compared with the scores of previous methods, are presented in the bar chart below 9

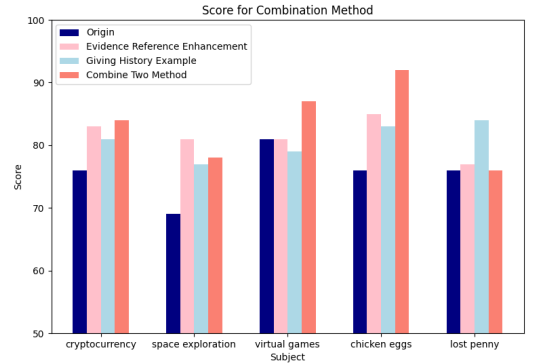


Figure 9: Score comparison between all methods.

According to the figure above, it is evident that the combination method nearly outperforms all other methods.

VII Conclusion

In conclusion, based on the analysis presented above, we propose the optimal configuration by setting the parameters as follows:

```
"model": "gpt-3.5-turbo-16k",
"frequency_penalty": 0.3,
"n": 1,
"presence_penalty": 0.3,
"top_p": 0.1
```

Furthermore, by requesting the reference of evidence and data, and providing agents with historical examples as templates, we have achieved the following performance shown in the histogram Fig. 10.

In addition we also provide the box plot 10 for a concise summary of the distribution of the results, including the median, quartiles.

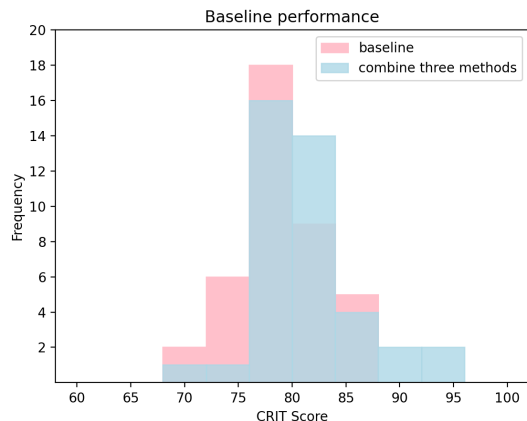


Figure 10: Baseline and improved score for all 40 topics.

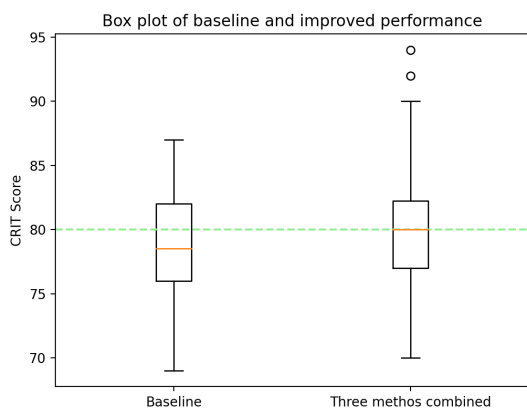


Figure 11: Baseline and improved score for all 40 public topics.

Additionally, in this experiment, we also employed the Mann-Whitney U test to compare two sets of data. The null hypothesis posited that the two populations are equal. Following the test, we obtained a p-value of

$$p - value = 0.0680 \quad (5)$$

Since this p-value is less than our chosen significance level, we have sufficient evidence to reject the null hypothesis. Therefore, we get the conclusion that there is a significant difference between the two populations, which means the modify exactly effect the final performance.

Table VI: Result of combine all methods vs. baseline.

	mean score	mediun score
baseline	78.65	78.5
combine	80.725	80

As shown by the figures, table and $p - value$, it's evident that our approach has indeed increased the CRIT score. The drawback of our approach is that it may restrict ChatGPT's conversational behavior or lead it to overly prioritize citations at the expense of highlighting truly crucial conclusions. Despite this, the advantages of our approach is that by let the agents to use references to substantiate information, thereby increasing the credibility of debate outcomes. Additionally, by exposing them to historical examples, we can effectively help them understand the expected format of conclusions, reducing the likelihood of missing information in the outputs.

Acknowledgments

Our collective contribution to this project has been equal, with each of us putting in significant effort. It is a widely held belief among us that our dedication and hard work deserves an equal grade for the project.

Appendix

Auto Program

Link:
https://github.com/camlet0630/2023HTML-bonus/blob/main/chatUI_auto

Complete Prompt

Link:
https://drive.google.com/file/d/14LZchL9uZ1bTzRJNXCDDe-6EKndL_3Bnb/view?usp=drive_link

[1] TAs, Propmt example.

[2] OpenAI, Api reference - openai api.

[3] E. Y. Chang, Socrasynth: Socratic synthesis for reasoning and decision making (2023).